

Cognitive factors in the perception of Spanish stress placement: implications for a model of speech perception*

TIMOTHY L. FACE

Abstract

Four cognitive factors potentially influencing the perception of Spanish stress placement are examined experimentally through a nonce word perception task, and the results are considered from the perspective of their implications for a psychological model of perception. The four factors considered for a potential influence on the perception of Spanish stress placement are segmentally similar words, syllable weight, subregularities in the lexicon, and morphological category. Each of these, with the exception of syllable weight, is shown to influence the perception of Spanish stress placement. The results of the experiment probe the structure of the lexicon in a way that shows that speakers are sensitive to lexical organization, and hence provide a window into the lexical system.

1. Introduction

Most experimental studies that have examined the nature of Spanish stress have focused principally on the acoustic correlates of stress (i.e., fundamental frequency, duration, and intensity) both in production studies (e.g., Navarro Tomás 1918; Quilis 1971, 1993; Simões 1996; Face 1999) and in perception studies (e.g., Enríquez et al. 1989; Llisterri et al. 2003). These acoustic correlates of stress, however, are only one aspect of a larger picture. Quilis (1971: 54) states that “the sensation of a physical stimulus is channeled through the structures of a language” (translation mine). While he does not address this issue any further, as his study focuses on the physical stimulus, Quilis indicates that there is a psychological aspect of perception through which listeners sense that they have perceived a physical stimulus. While factors, though primarily syllable weight, other than the three acoustic correlates of

stress have been considered in a handful of recent studies (Aske 1990; Alvord 2003; Bárkányi 2002; Eddington 2000, 2004; Face 2000; Face and Alvord 2005; Waltermire 2004), of these only Face (2000) considers the perception of stress placement. Even in that study, however, the acoustic correlates of stress were not adequately controlled and influenced the experimental results (see Section 2.2.1 for further discussion).

In the present study, four cognitive factors potentially influencing Spanish stress perception are examined experimentally through a nonce word perception task that probes the structure of the lexicon, and the results are considered in terms of their implications for a psychological model of speech perception. I refer to the factors considered as ‘cognitive factors’ because they are factors other than the acoustic cues to stress, yet they may lead listeners to perceive one of the syllables of the word as bearing stress. These four factors, each examined for their influence on the perception of stress placement in nonce words, are 1) the stress pattern of a known word similar phonetically and phonologically to the nonce word, 2) syllable weight, 3) a subregularity in the lexicon, and 4) the morphological category of the word.

Moving now from the experimental component of the present article to the theoretical, models of speech perception generally model lexical access of existing words. The present study’s use of nonce words adds a different perspective to the consideration of models of speech perception. Speaker/hearers also encounter words that are not recognized as an existing part of the lexicon, such as a previously unknown word to the hearer, a word not immediately recognized due to poor pronunciation or other factors, a newly coined word, a foreign borrowing, or an unfamiliar proper name. While these cases are certainly a small portion of the words processed by a listener, they are nonetheless encountered in real life situations and processed. Because of this, it is of interest to determine what factors enter into perception in these cases and what light these might shed on the process of speech perception.

As a vehicle to addressing the perception of words that are not recognized as an existing item in the lexicon, the present study examines four factors (mentioned above) that are hypothesized to influence the perception of Spanish stress. The Spanish stress system produces numerous minimal pairs (and triplets) differing only in the placement of stress within the word.¹ In addition, the location of stress within the word has no effect on vowel quality (Quilis and Esgueva 1983).² Example minimal pairs and triplets of Spanish words differing only in the location of stress are given in (1), where stress location is marked whether required or not by Spanish orthographical standards.

(1)	a.	depósito 'deposit', n.	deposító 'deposit', 1st sg. pres. indic.	deposító 'deposit', 3rd sg. past indic.
	b.	número 'number', n.	numéro 'number', 1st sg. pres. indic.	numeró 'number', 3rd sg. past indic.
	c.	estímulo 'stimulus', n.	estimúlo 'stimulate', 1st sg. pres. indic.	estimuló 'stimulate', 3rd sg. past indic.
	d.	mentíra 'lie', n.	mentirá 'lie', 3rd sg. fut.	
	e.	háblo 'talk', 1st sg. pres. indic.	habló 'talk', 3rd sg. past indic.	
	f.	comprára 'buy', 1st/3rd sg. past subj.	comprará 'buy', 3rd sg. fut.	

In spite of the existence of numerous minimal pairs and triplets, distinguished only by stress placement, stress placement in Spanish is highly predictable. A discussion of stress placement in Spanish, including both distributional facts and results of previous experimental studies, is presented in Section 2. Section 3 details the experimental methods used in the present study. In Section 4 the experimental results of each part of the experiment are reported and discussed. Section 5 contains a general discussion of the overall experimental results as well as their implications for a psychological model of speech perception. The conclusions of the present study are presented in Section 6.

2. Stress placement in Spanish

2.1. *Distributional facts*

As noted above, Spanish stress placement is not completely predictable based on phonological information, and numerous minimal pairs and triplets, differing only in stress placement, exist. Yet stress placement is highly predictable, both in verbs and nonverbs, though the predictability is based on different information in the two cases. In both verbs and nonverbs, stress may fall on any of the last three syllables. In regular verbs, stress placement is completely predictable based on verb tense, mood, person, and number. In the majority of cases, verb stress falls on the

Table 1. *Stress placement on the 4829 most frequent polysyllabic words in Spanish*

Word ending	Final stress	Penult stress	Antepenult stress
Vowel	178	2494	178
/s/	20	909	94
Consonant (except /s/)	778	176	2

penultimate syllable, but there are certain forms where it falls on the final syllable (e.g., the infinitive, first and third person singular preterit forms, all future forms except for first person plural) or on the antepenultimate syllable (e.g., first person plural in the imperfect indicative, imperfect subjunctive, and conditional). While not completely predictable as it is in regular verbs, stress placement in nonverbs is also highly predicable. In general, nonverbs (and, to a lesser extent, verbs) that end in a vowel or /s/ have penultimate stress, while those ending in a consonant other than /s/ have final stress, and these patterns may be considered unmarked for Spanish, with other patterns being marked. When the 4829 most frequent words in Spanish, taken from the Alameda and Cuetos (1995) frequency dictionary, are considered, these generalizations about stress placement produce 87% accuracy in predicting stress placement, and this level of accuracy is in spite of these words including both regular and irregular verb forms as well as nonverbs. These results can be seen in Table 1, adapted from Eddington (2000: 96).

2.2. *Experimental studies*

Besides the general patterns of stress placement in Spanish, it is pertinent here to consider experimental work on stress placement in Spanish and the factors affecting it. There exists experimental work on three of the four factors to be considered in the present study: syllable weight, sub-regularities in the lexicon, and morphological category. Of these factors, syllable weight is the one that has received the most attention in experimental studies, and I begin the discussion with this work.

2.2.1. *Syllable weight.* Whether or not Spanish has a quantity sensitive stress system is a matter of debate. While the Spanish stress system developed from the clearly quantity sensitive stress system of Latin, theoretical analyses do not agree as to whether modern day Spanish is quantity sensitive. Experimental studies have begun to look at this issue from different perspectives. Face (2000) considered the effects of syllable weight on the perception of Spanish stress placement. Bisyllabic and trisyllabic nonce

words were created with every possible combination of heavy and light syllables, and were synthesized to maintain a flat and unvarying F0 and intensity, as well as to give all vowels equal duration. The synthesized nonce words were presented to subjects who were asked to indicate which syllable carried stress.

Face (2000) found that there was a strong preference for the unmarked stress pattern (i.e., penultimate stress in words with a light, or open, final syllable and final stress in words with a heavy, or closed, final syllable) to be perceived in the nonce words. The overall percentage of perception of the unmarked stress pattern was 74.2%, with a percentage of 68% in words with a heavy final syllable and 80.3% in words with a light final syllable. The weight of the final syllable was highly significant in determining perceived stress. When the final syllable was heavy, final stress was generally perceived, and when the final syllable was light, penultimate stress was generally perceived. This led Face (2000) to claim that syllable weight causes perception of unmarked stress. With respect to the higher percentage of perception of the most common stress pattern when the final syllable is light, Face (2000) argues that while phonological structure may make either the penultimate syllable or the final syllable unmarked (when the final syllable is light or heavy, respectively), penultimate stress is a truly default stress pattern. This is supported by looking at cases where the phonologically unmarked stress pattern is not perceived. In cases where marked stress was perceived, in nearly 50% of the cases penultimate stress was perceived, with the remaining cases split fairly evenly between the final and antepenultimate syllables. In addition, in control words that did have acoustic prominence on one particular syllable, nearly two-thirds of the subjects' errors "corrected" the acoustic stress to the unmarked stress pattern. When acoustic stress corresponded with phonologically unmarked stress, it was perceived correctly nearly 100% of the time.

Looking beyond unmarked stress (based on the weight of the final syllable of the word), Face (2000) notes that the rightmost heavy syllable, regardless of whether or not it is the final syllable of the word, seems to attract perceived stress. Examining those words with a light final syllable, he shows that the weight of the penultimate syllable is a factor in the perception of stress. When the penultimate syllable is heavy, it is perceived as stressed 86% of the time, and the antepenultimate syllable is perceived as stressed only 3% of the time. When both the final and the penultimate syllables are light, the penultimate syllable is perceived as stressed 66% of the time, and the antepenultimate syllable is perceived as stressed 18% of the time. The heavy penultimate syllable attracts stress and all but prevents perception of antepenultimate stress. In addition,

when both the final and penultimate syllables are light, the antepenultimate is significantly more likely to be perceived as stressed when it is heavy than when it is light.

Face (2000) takes the results of his experiment as evidence that syllable weight influences the perception of Spanish stress. Furthermore, he states that the results provide evidence that “syllable weight is not just an abstract issue to be dealt with by phonological theory, but that it is a very real cognitive factor that influences stress perception even beyond determining unmarked stress” (11). While the conclusions drawn by Face (2000) appear well-motivated by his experimental results, there is one potentially confounding problem with the methodology. As mentioned above, all vowels were of equal duration. This means that a coda consonant added to make a syllable heavy also increased the duration of the syllable, making heavy syllables longer than light syllables. While this is an accurate representation of what happens in real Spanish speech, for the purpose of the experiment it means that there could have been an acoustic correlate of stress leading to stress perception. Therefore, it could be that heavy syllables were perceived as stressed not because they were phonologically heavy, but because they were longer than light syllables. The present study corrects this error of experimental design in reconsidering whether syllable weight really is a factor in the perception of Spanish stress or whether Face’s (2000) results were skewed by the extra duration of heavy syllables.

Bárkányi (2002) presents results from another type of experimental study, and her results differ considerably from those of Face (2000). Bárkányi presents nonce words to subjects orthographically, and asks them to mark which syllable they would stress if producing the nonce words. For words with a heavy penultimate syllable, subjects do not avoid marking antepenultimate stress as would be expected if Spanish were quantity sensitive. Rather antepenultimate stress is selected by subjects in these cases from 12.5% to 30% of the time, depending on the specific CV structure of the word. Bárkányi concludes that stress placement is not dependent on syllable structures contained in the word, at least not in the way that has commonly been proposed in theoretical analyses of Spanish stress that take into account syllable structure (Harris 1983, 1992; Núñez Cedeño 1988; Lipski 1997).

Waltermire (2004) also presents nonce words to subjects orthographically, and asks them to mark which syllable they would stress if producing the nonce words. His findings indicate, like those of Face but contrary to those of Bárkányi, that syllable weight has a significant effect on stress placement. His results differ from Face’s, though, in that they show a much more complex interaction of the weights of all syllables in subjects’

selection of stress placement. Where Face found only the weight of the final syllable to be significant in all cases, Waltermire reports that the weights of all syllables are significant in determining stress placement in some way, and often the best predictor of stress placement is the combination of multiple syllable weights. Waltermire (2004: 184) speculates that the differences between his and Face's results may be due to the different tasks the subjects performed, stating that "when faced with determining the most logical stress for a nonce word at the very moment of speech, it seems likely that speakers would choose the most frequently occurring stress positions (final or penultimate) while not fully considering the least frequently occurring one (antepenultimate)", as may be done in responding to a written questionnaire. Interestingly, Waltermire does not offer any commentary on the differences between the results of his study and that of Bárkányi.

A third study employing a written questionnaire was carried out by Alvord (2003). His study differs from those of Bárkányi (2002) and Waltermire (2004) in that rather than ask subjects to mark stress on nonce words, he presents subjects with orthographic representations of nonce words including an orthographic accent mark to indicate the location of stress. Subjects were then asked to judge each nonce word as either a possible or impossible Spanish word. Nonce words with a heavy penultimate syllable and antepenultimate stress were overwhelmingly accepted (94%) as possible Spanish words. Based on these results, Alvord not only concluded that, in agreement with Bárkányi, that Spanish is not quantity sensitive, but also went a step further. He questioned whether the oft cited restriction against Spanish words with a heavy penultimate syllable and antepenultimate stress (e.g., Harris 1983, 1992; Lipski 1997) might not be productive restrictions at all (regardless of whether the restriction is due to quantity sensitivity or not), but rather the result of historical developments in the language (cf. Roca 1990).

Face and Alvord (2005) set out to investigate two purported restrictions on Spanish stress placement, including the restriction against words with a heavy penultimate syllable and antepenultimate stress, using a perception task.³ With respect to the aforementioned restriction, this allowed investigation of whether the high acceptance rate in Alvord (2003) was the result of the methodology or whether the results would be similar when subjects were presented with nonce words auditorily. As in Alvord (2003), subjects were asked to determine whether each nonce word was a possible or impossible Spanish word. While the rate of acceptance of words with a heavy penultimate syllable and antepenultimate stress is not as high as in Alvord (2003), they were still accepted 67% of the time. This is especially impressive when compared to a control group of words

expected to be impossible because of violations of Spanish phonotactic constraints, where the acceptance rate was only 11%. This result is taken as evidence against Spanish being quantity sensitive and in favor of Roca (1990) and Alvord's (2003) claim that there is not a productive restriction against words with a heavy penultimate syllable and antepenultimate stress.

Eddington (2004) takes another approach to investigating the effect of syllable weight on stress placement. In a follow-up to Eddington (2000), which did not consider syllable weight, Eddington (2004) uses Analogical Modeling of Language (AML; Skousen 1989, 1992, 1995) to determine whether syllable weight and other factors are able to accurately predict stress placement in Spanish by analogy. This study is different from those previously discussed in that no subjects are involved. Rather the 4970 most common words in Spanish, based on the Alameda and Cuetos (1995) frequency dictionary, are employed and coded with various types of information, including phonemic representation, CV structure, and syllable weights. These words are divided into ten groups of 497 words, and for each simulation nine of the groups serve as the lexicon while the tenth is the test set for which the AML algorithm predicts stress placement based on the patterns in the lexicon. Eddington's results show that phonemic representation predicts stress placement significantly better than do CV structure or syllable weights alone. Only when some phonemic representation is added to CV structure or syllable weight representations do the results rival those of phonemic representation alone. Nonetheless, Eddington is quick to point out that syllable weight should not be discounted as a factor in stress placement, even if it may not be the best predictor of stress placement. When a set of nonce words were presented to the AML algorithm, with the real words serving as the lexicon, the most accurate results were produced when the variable taken into account by AML included the syllable weights of the final three syllables and the final phoneme of the words. Eddington concludes, then, that abstract entities such as syllable weight may play a role in stress placement, though the specifics of that role are left for future studies.

The sum of the experimental studies on the effects of syllable weight on Spanish stress placement is that there is mixed support. Most studies seem to point to Spanish not being quantity sensitive and not actively prohibiting words with a heavy penultimate syllable from having antepenultimate stress. Nonetheless, syllable weight does factor into stress placement in one way or another in a number of studies. While syllable weight is by far the most studied factor on Spanish stress placement in experimental studies, there appears to be the need for more research to determine exactly what role it plays.

2.2.2. *Subregularities in the lexicon.* The only experimental study of a lexical subregularity in Spanish stress placement was carried out by Aske (1990). Aske conducted an experiment to test the psychological reality of general stress rules in Spanish. In doing so, one of the issues he considered was the presence of a subregularity in the lexicon. Specifically, he noted that while the general (and overwhelming) pattern is for nonverbs ending in [n] to have final stress, nonverbs ending in [en] are split fairly evenly between penultimate and final stress. Aske pointed out that this is a good set of words to use in a test of the psychological reality of Spanish stress rules. If a general stress rule is employed by speakers, a new word ending in [n] should be produced with final stress. If, on the other hand, stress is stored in the lexicon, speakers should access the lexicon and use the surface patterns found there in assigning stress. If this is the case, new nonverbs ending in [en] should be produced with a mix of penultimate and final stress, reflecting the distribution of stress in words in the lexicon ending in [en]. All other new nonverbs ending in [n] where the vowel immediately preceding the [n] is any vowel except [e] should be produced almost exclusively with final stress, again representing the distribution of stress in words in the lexicon that end in these segmental strings.

Aske (1990) created 12 nonce words ending in [n], with six preceded by [e] and the other six preceded by other vowels. These were then embedded in contextualizing sentences so that each was interpreted as either a noun or an adjective. Subjects read the sentences, which were written in all capital letters since orthographic accent marks are commonly not marked on capital letters in Spanish. The results were that for words ending in [en], final stress was produced 55.6% of the time and penultimate stress was produced 43.5% of the time. In contrast, words ending in an [n] preceded by any vowel except [e] were produced with final stress 96.8% of the time and penultimate stress only 3.2% of the time. Aske (1990) argues that these results support a model in which speakers access actual patterns in the lexicon and that they are not compatible with a model in which general stress rules are employed. In addition, he notes that an attempt at incorporation of these results into a rule-based account would be problematic since the subregularity (i.e., a fairly equal distribution of penultimate stress and final stress in words ending in [en]) is variable rather than categorical.

2.2.3. *Morphological category.* In addition to the subregularity of penultimate stress on words ending in [en], Aske (1990) considers another difference in stress placement based on the morphological category of the word. He notes that the majority of words ending in [iko] or [ika] have antepenultimate stress due to the unstressed nature of the adjective-forming

suffix [ik], not considering diminutives with this ending (e.g., *papelico* ‘paper; dim.’, *matica* ‘blade; dim.’), which are productively formed and thus not listed in standard dictionaries. Of the words ending in [iko] or [ika] that have antepenultimate stress, 76% are adjectives (e.g., *fanática* ‘fanatical’), and another 19% are both adjectives and nouns (e.g., *química* ‘chemistry n., chemical adj.’). Only 5% are nouns only (e.g., *república* ‘republic’). Of the adjectives, nearly half are not analyzable, or only partially analyzable, as consisting of a stem plus [iko] or [ika]. Aske (1990) predicts that if speakers follow a general stress rule, they will produce new words in [iko] or [ika] with penultimate stress since there is not evidence for a suffix, and therefore morphological category will not be relevant for stress assignment. On the other hand, he predicts that if speakers consult the lexicon and search for similar words, morphological category may be one of the factors used in determining stress placement.

Aske (1990) created 12 nonce words ending in [iko] or [ika] and placed them into sentences so that six were only interpretable as nouns and six were only interpretable as adjectives. As with the study of words ending in [n], the sentences were written in capital letters, since orthographic accent marks are often not used on capital letters. The results were that for the adjectives antepenultimate stress was produced 83.3% of the time and penultimate stress 16.7% of the time. For the nouns antepenultimate stress was produced 63.4% of the time and penultimate stress 36.6% of the time. While these results point in the direction of speakers treating nouns and adjectives ending in [iko] and [ika] differently, the results are not statistically significant, and there is still a very large percentage of antepenultimate stress produced on nouns. While the results are much less convincing than those found for words ending in [n], Aske (1990) notes that the results do show that speakers treat words ending in [iko] and [ika] differently from other words ending in a vowel, and that this supports a model where speakers consult existing patterns in the lexicon.

Aske’s (1990) results for words ending in [iko] or [ika] do not provide sufficient motivation that morphological category is a factor in stress placement. However, this distinction between antepenultimate and penultimate stress in words ending in [iko] and [ika] may not be a very robust distinction, especially as determined by morphological category. Using Bosque and Pérez Fernández’s (1987) reverse dictionary, he finds a total of 3580 words ending in [iko] or [ika] with antepenultimate stress and only 121 with penultimate stress (of which only 19 he considers to be familiar to him). Of those with antepenultimate stress, he finds that about 19% are both nouns and adjectives, and only 5% are nouns only. That means that 24% can be nouns. 24% of the 3580 words with antepenultimate stress is 859. That means that 859 nouns have antepenultimate stress

while only 121 have penultimate stress. So it should have been expected that nouns would also be produced with antepenultimate stress in the majority of cases. While Aske's (1990) results support his claim that speakers access the lexicon rather than general stress rules, they offer no insight as to whether or not morphological category is one of the factors used in determining stress placement. Interestingly, while Eddington's (2004) study set out primarily to examine phonological factors in stress placement, he includes morphological information in some AML simulations of Spanish stress placement and finds that morphological variables are among the most influential in determining stress placement. It is yet to be seen, however, whether actual human performance data support Eddington's finding.

3. Experimental methodology

3.1. Nonce word task

In order to examine whether the four above-mentioned factors influence the perception of Spanish stress placement, an experiment was designed in which trisyllabic nonce words, following the phonotactic patterns of Spanish, were created and synthesized. The synthesized nonce words were then presented to subjects who heard the synthesized nonce words and then indicated for each the syllable that they believed to be stressed. The hypothesis was that subjects would make comparisons of the stimulus to patterns of different types (within words, across words, etc.) stored in the lexicon and use this information in the perception of stress placement. The nonce words used to examine each of the issues considered in the present study were combined and randomized, with the exception of those testing the influence of morphological category. In order to test the influence of morphological category, a sentence context, rather than a single word in isolation, was required, and therefore this portion of the experiment was conducted separately, albeit with the same subjects and during the same testing session. The principles guiding the creation of the nonce words varied based on the specific issue they were meant to test, and these are discussed in the following section.

Once the nonce words to be used in the study (and their corresponding contexts in the case of those testing the influence of morphological category) were created, they were synthesized using the MBROLI speech synthesizer (Dutoit et al. 1996) with a Spanish voice.⁴ The use of this synthesizer allowed for the fundamental frequency to be held steady throughout the production of the nonce word and for the duration of each

syllable to be identical so as to prevent either of these acoustic correlates of stress from affecting the decision of the subjects. While the MBROLI speech synthesizer does not allow for control of intensity, an investigation of each synthesized nonce words found that intensity was consistent throughout each word. Therefore, like fundamental frequency and duration, intensity could not serve as an acoustic cue to stress in this experiment. This removal of acoustic cues to stress is of utmost importance in the present study so that the results cannot be attributed to the perception of acoustic cues. Rather the perception of stress placement must be due to factors other than acoustic cues to stress, such as the factors investigated in the present study (i.e., stress pattern of a similar real word, syllable weight, lexical subregularity, and morphological category).⁵ This also means that the experimental results will have larger implications for models of speech perception since factors such as these have not yet been incorporated into such models.

For each part of the experiment (i.e., words in isolation and sentences), there was a practice session so that the subjects could become acclimated to the synthetic voice and to the task of selecting the syllable of the nonce word that was perceived as stressed. The recording of the perceived stressed syllable was done on paper. The paper answer sheet consisted of an item number (i.e., 1 for the first nonce word presented, 2 for the second, etc.) followed by the numbers 1, 2 and 3. The subject would circle 1 if the first syllable of the word was perceived as stressed, 2 if the second was perceived as stressed, and 3 if the third syllable was perceived as stressed. Interspersed with the test words in both parts of the experiment were additional cases where the nonce word did have an acoustically stressed syllable. This was done as a control to be sure that when an acoustically stressed syllable was present, the subjects could accurately identify it. Any potential subject who did not accurately identify the acoustically stressed syllable at least 90% of the time was excluded from the study.⁶

3.2. *Nonce words*

3.2.1. *Similar real words.* In order to test whether the stress pattern of a similar word has an effect on the perception of Spanish stress placement in words not recognized as part of the lexicon, ten real trisyllabic Spanish words ending in a consonant and having final stress (e.g., *hospital* [ospitál] 'hospital') were chosen. These were the similar words that could have an influence on the perception of stress placement in a segmentally similar nonce word. From each of the real words, three nonce words were

created that were segmentally identical to the real word through the penultimate segment. Two of the three nonce words simply lacked the final consonant of the real word, while in the third the original final consonant was replaced by [s] (see Appendix A for the complete list of nonce words). Since the nonce words are vowel-final and [s]-final, Spanish stress generalizations indicate that they should have penultimate stress, and, if the similar real word is not an influence, this should be what listeners perceive. However, if the similar real word does influence the perception of stress placement, then the nonce words should generally be perceived as having final stress, since the similar real words all have final stress. Care was taken to ensure that the nonce word was not the beginning of multiple real words, since this could lead to the conflicting influence of different stress patterns from different real words.⁷

The two vowel-final nonce words differed in the quality of the final vowel. In one case the quality was that of the corresponding vowel in the real word (where it preceded the final consonant), and in the other it had the quality of a word-final vowel (i.e., the quality of the vowel in a word-final open syllable, which is more open than when preceded by a coda consonant). The three nonce words were created in this fashion in order to create different degrees of phonetic similarity to the real word to determine if there was a gradient difference in stress perception based on the degree of similarity. The vowel-final nonce word in which the final vowel has the same quality as the vowel in the original word, where it is colored by the following consonant, is phonetically more similar to the real word than the vowel-final nonce word where the final vowel has the more open quality of a word-final vowel. This difference in vowel quality was included in order to determine whether this phonetic difference, in spite of being two instances of the same phonological vowel, would affect subjects' use of analogy with the real word. If so, we would expect more selection of final stress (i.e., the pattern in the real word) in the case where the final vowel of the nonce word has the quality of the vowel in the real word. If not, we would expect similar percentages of perceived stress placement in both of the vowel-final nonce words.

3.2.2. Syllable weight. Forty nonce words were created to test the effect of syllable weight on the perception of Spanish stress (see Appendix B). These forty words are all trisyllabic and include all possible combinations of heavy and light syllables. Five nonce words were created for each of the eight possible combinations. Correcting the potential methodological problem of Face (2000), the durations of all syllables (rather than vowels) in each word were identical. Therefore in heavy syllables, the vowel was shortened in order to accommodate the coda consonant.⁸ In

addition, it is essential that the nonce words designed to test the effect of syllable weight on Spanish stress not resemble any real Spanish words, as the stress pattern of a real word may influence the perception of stress, as will be seen in Section 4.1. For this reason, a preliminary set of synthesized nonce words was reviewed by a native Spanish speaker, and any that sounded similar to a real word were subsequently replaced.

3.2.3. *Subregularity in the lexicon.* Nonce words were designed to test the subregularity noted by Aske (1990) that nonverbs ending in [n] overwhelmingly have final stress, except for those where the vowel preceding the [n] is [e]. In this latter case there is a fairly even split between penultimate and final stress. Three sets of nonce words were created, with each set containing ten nonce words (see Appendix C). One set contained words ending in [an], one set words ending in [en], and one set words ending in [on]. While Aske (1990) makes a distinction between words ending in [en] and all other words ending in [n], the present study required one further distinction due to the nature of the experimental design. Aske (1990) had all of the words contextualized in sentences where the only interpretation was as a noun. However, since the present study presented individual nonce words in isolation to the subjects, it would be possible for words ending in [en] to be interpreted as the third person plural present tense form of a verb, and these forms always have penultimate stress. Therefore a higher percentage of perceived penultimate stress could be the result of the existence of verb forms with this stress pattern, in contrast to forms ending in [on], where the nonexistence of present tense verb forms ending in this segmental sequence prohibits a verbal influence toward perceived penultimate stress.⁹ The [an] and [on] sets were both included so that the influence of verb forms could be noted. Since the third person plural present tense form of first conjugation verbs ends in [an], comparing this set with the [en] set will show whether the [en] set leads to perceived penultimate stress more than the [an] set. If so, then the lexical subregularity is likely the reason. On the other hand, a close similarity between the [en] and [an] set, but with both having a higher percentage of perceived penultimate stress than the [on] set, would indicate that verb forms, rather than a lexical subregularity, explain the perceived penultimate stress in the [en] forms.

3.2.4. *Morphological category.* In order for the present study to test whether or not morphological category affects the perception of Spanish stress, a salient stress difference between words in different morphological categories, but sharing the same terminal segmental string, had to be found. Such a difference exists between nouns and third person plural

present tense verb forms ending in [n]. Nouns ending in [n] generally have final stress, and the third person plural present tense verb forms always have penultimate stress (except when monosyllabic). These verb forms end in either [an] or [en], but forms ending in [an] were chosen for the present study. In this way it could be assured that there would be a clear difference between stress patterns, since nouns ending in [en] often have penultimate stress.

The 10 nonce words ending in [an] that were used to test the effect of a lexical subregularity on the perception of Spanish stress placement were used to test the effect of morphological category on the perception of Spanish stress placement as well. Each of these ten nonce words was inserted into two contextualizing sentences. In one sentence (i.e., *María quiere un _____*. 'María wants a _____'), the nonce word could only be interpreted as a noun, and in the other (i.e., *Los alumnos _____ en la clase*. 'The students _____ in class.') it could only be interpreted as a verb. The contextualizing sentence was synthesized to sound as natural as possible, while the nonce word inserted, like the other nonce words in the present study, had no syllable that was acoustically more prominent than the others.

If morphological category does influence the perception of Spanish stress, then those cases in which the nonce words can only be interpreted as a noun should show a strong preference for perceived final stress, and those cases in which the nonce word can only be interpreted as a verb should show a strong preference for perceived penultimate stress. Since this distinction based on morphological category is so extreme in real words, it is expected that the results of the present study should be quite clear if indeed morphological category has an effect on the perception of Spanish stress. Any results that do not clearly show a difference in perceived stress based on morphological category would likely mean that morphological category does not have an effect on Spanish stress perception. In providing instructions to subjects, no mention was made of morphology. They were simply told to listen now for the stress pattern of the nonce word embedded in a sentence. Two additional contextualizing sentences embedded with nonce words served as foils.

3.3. Subjects

The subjects who participated in the experiment were 10 native Spanish-speaking graduate students at the University of Minnesota who had not lived in the United States prior to attending graduate school. All grew up in a monolingual environment, though they came from various parts

of the Spanish-speaking world. While the same three acoustic correlates of stress (i.e., pitch, duration, and intensity) are used in different varieties of Spanish, there may be dialectal differences in the importance of each. In the present study, however, this is not a factor since the experiment neutralizes these acoustic correlates of stress in order to test the role of structural factors on stress perception. The issues addressed in the present study apply equally to all varieties of Spanish.¹⁰ All of the subjects were naïve with respect to the purpose of the study and had no training in phonetics or phonology.

4. Results

4.1. Similar words

The results for the nonce words testing the effect of a similar word on the perception of stress placement are given in Table 2.

These results show a significant difference in distribution of perceived stress placement according to the nature of the final segment of the nonce word. More specifically Table 2 shows that vowel-final words are perceived as having final stress more often than penultimate and antepenultimate stress combined, indicating an analogical affect of the stress pattern of the real word on perception of these nonce words. However, there is a considerable difference in the distribution of perceived stress between the two categories of vowel-final nonce words. This indicates that the degree of phonetic similarity to the real word also plays a role in perception. That is, not only is the phonemic nature of the vowel important in perception, but different phonetic qualities of the same phonemic vowel will lead to different degrees of analogical influence. With respect to the [s]-final nonce words, penultimate stress was perceived more often than final stress, indicating less influence of the stress pattern of the similar real word than is the case for the vowel-final nonce words. Nonetheless, it is

Table 2. Perceived stress placement in nonce words with a similar real word

Perceived stressed syllable	Vowel-final (original vowel quality)	Vowel-final (quality of word-final vowel)	[s]-final
Antepenultimate	12	12	18
Penultimate	16	29	45
Final	72	59	37
Total	100	100	100

Chi-square: $\chi^2(4) = 27.74$, $p < 0.0001$

worth noting that the 37% of perceived final stress in these cases is higher than might be expected if there were no influence of the similar real word. In words predicted to have penultimate stress, Face (2000) reports the perception of final stress in only 13.5% of the tokens. Based on this, the 37% of perceived final stress for [s]-final words in Table 2 seems to demonstrate that the stress pattern of the real word still has some influence on the perception of stress in the [s]-final nonce words.

4.2. Syllable weight

In order to begin the investigation of the role of syllable weight in the perception of Spanish stress, I consider the effect of the weight of the final syllable, as it is this that determines the phonologically unmarked stress pattern. The results are shown in the leftmost columns of Table 3.

These results confirm the finding in Face (2000) that the weight of the final syllable is a factor in the perception of Spanish stress. When the final syllable is heavy, it is generally perceived as stressed. When the final syllable is light, the penultimate syllable is generally perceived as stressed. But the overall percentage of unmarked stress perception is 61.8%, which is notably lower than the 74.2% reported in Face (2000). Also of interest is that the unmarked stress pattern is perceived more often in consonant-final words (i.e., final stress is unmarked) than in vowel-final words (i.e., penultimate stress is unmarked). Combining both categories, final stress is that which is most often perceived. Nonetheless, when the categories are separated it is clear that the phonologically unmarked stress pattern (i.e., penultimate stress when the final syllable is light, and final stress when it is heavy) is still the most common pattern perceived in the present study.

Table 3. *Perceived stress placement by syllable weights*

Perceived stressed syllable	Light final syllable	Heavy final syllable	<i>Light final syllable</i>		<i>Light final and penult. syllables</i>	
			Light penult. syllable	Heavy penult. syllable	Light antepenult. syllable	Heavy antepenult. syllable
Antepenult.	18	22	11	7	6	5
Penult.	112	43	57	55	29	28
Final	70	135	32	38	15	17
Total	200	200	100	100	50	50
Chi-square	$\chi^2(2) = 53.19$, p < 0.0001		$\chi^2(2) = 1.45$, p = 0.49		$\chi^2(2) = 0.23$, p = 0.89	

I now consider whether heavy syllables attract perceived stress, as claimed by Face (2000). That is, when the final syllable is light, is penultimate stress perceived more often when the penultimate syllable is heavy? When both the final and the penultimate syllables are light, is antepenultimate stress perceived more often when the antepenultimate syllable is heavy? The results for words with a light final syllable are given in the middle columns of Table 3, while those for words with both a light final and a light penultimate syllable are given in the rightmost columns.

The results presented in Table 3 for the effects of the weight of the penultimate syllable when the final syllable is light and for the effects of the weight of the antepenultimate syllable when the final and penultimate syllables are light go against the claim made by Face (2000) that a heavy syllable attracts perceived stress in Spanish. When the final syllable is light there is very little difference between the percentage of perceived penultimate stress based on the weight of the penultimate syllable. Similarly, when both the final and penultimate syllables are light there is almost no difference between the percentage of perceived antepenultimate stress based on the weight of the penultimate syllable. In fact, while the difference is by no means significant, in both cases a light syllable is perceived as stressed more often than a heavy syllable. Since a heavy syllable does not attract perceived stress in these cases, it brings into question the role of syllable weight in Spanish stress perception. This perceptual evidence supports other recent experimental evidence that Spanish is not quantity sensitive (Alvord 2003, Bárkányi 2002).¹¹

But if Spanish is not quantity sensitive, how can we account for the fact that the weight of the final syllable seems to be a relatively strong factor in perceived stress placement? For Spanish, a syllable may be considered heavy if it ends in a consonant and light if it ends in a vowel, as Spanish does not have a phonological vowel length distinction. I propose, then, that the most common pattern of perceived stress is not the result of syllable weight (as there is evidence against the role of syllable weight as a factor in the perception of Spanish stress as discussed above), but rather is the result of the nature of the final segment of the word. Speaker/hearers' lexical knowledge that the majority of words ending in a vowel have penultimate stress while the majority of words ending in a consonant have final stress is accessible during language use and can be used actively in production and perception. Since it is segmental information at the end of the word that determines the most common stress patterns, this explanation also accounts for why coda consonants in non-final syllables do not affect stress perception, a fact that could not be accounted for by a syllable weight explanation.¹²

4.3. Subregularity in the lexicon

The results for the nonce words testing the effect of a lexical subregularity on stress perception are given in Table 4.

The two sets (i.e., [en] and [an]) where the nonce words could be interpreted as a verb form have a higher percentage of perceived penultimate stress (and a corresponding lower percentage of perceived final stress) than the set (i.e., [on]) where the nonce words could not be interpreted as a verb form.¹³ Since the [an] set, as well as the [en] set, has a higher percentage of perceived penultimate stress than the [on] set, it appears that the possibility of these forms being verbs did have some effect on stress perception, and a Chi-square test shows a significant difference between [on] words and non-[on] words when [an]-words and [en]-words are combined ($\chi^2(2) = 6.54$, $p < 0.05$). While this pattern is of interest, it is also interesting that there is a notable difference in perceived stress between the [en] and [an] sets, indicating that being a possible verb form is not the only explanation needed. Notably, it is the [en] set that has a higher percentage of perceived penultimate stress (though a comparison of only the [en] set and the [an] shows a difference that is not statistically significant: $\chi^2(2) = 2.78$, $p = 0.25$). The difference between the [en] and [an] sets, then, could be the result of the lexical subregularity discussed by Aske (1990) — nouns ending in [en] are largely split between penultimate and final stress, while other nouns ending in [n] typically have final stress.

The results obtained in the present study are not as clear as those obtained by Aske (1990) with regard to the influence of a lexical subregularity on stress, but nonetheless indicate that the lexical subregularity is a factor in the perception of Spanish stress placement. The question could be raised as to why the results presented here are not as clear as Aske's (1990) results for production. A definitive answer cannot be given, though three potential explanations can be mentioned. The first is a simple accidental difference in distribution of stress placement between the two studies, with no linguistic explanation possible. While this is certainly possible, it seems preferable to offer a linguistic explanation if one is

Table 4. *Perceived stress placement in nonce words ending in [n]*

Perceived stressed syllable	[en] words	[an] words	[on] words
Antepenultimate	12	13	13
Penultimate	29	19	12
Final	59	68	75
Total	100	100	100

Chi-square: $\chi^2(4) = 9.32$, $p = 0.05$

available. A second potential explanation is that the results may have been stronger in the present study had the words been contextualized in sentences, assuring a nonverb reading of each. A third potential explanation is that Aske's (1990) experiment was one of language production while the one conducted for the present study is one of perception. There is no basis at the present time to claim that production and perception use the lexicon and the patterns found within it in different ways. In fact, I would argue that the lexicon plays a similar role in both production and perception. However, it could be that the pattern in question here plays a more active role in production than in perception. The question of whether production and perception make different degrees of use of (some) patterns found in the lexicon, however, must be left for future study. For the purposes of the present study, what matters is that the lexical subregularity does have an influence on the perception of Spanish stress placement

4.4. *Morphological category*

The results for the nonce words testing the effect of morphological category on stress perception are given in Table 5.

These results show a highly significant and very convincing difference between perceived stress on nouns and on verbs. Nouns ending in [an] are expected to have final stress, and final stress is perceived 81% of the time. Third person plural present tense verb forms always have penultimate stress when they are polysyllabic, and penultimate stress is perceived 86% of the time. These results show quite clearly that morphological category does have a very significant effect on the perception of Spanish stress placement. It can be assumed that Aske's (1990) results would have also shown a convincing effect of morphological category had he chosen a more robust morphological distinction. The results presented here also provide concrete performance data that support the finding of Eddington's (2004) computer simulation of stress assignment, where he finds

Table 5. *Perceived stress placement by morphological category*

Perceived stressed syllable	Nouns	Verbs
Antepenultimate	4	3
Penultimate	15	86
Final	81	11
Total	100	100

Chi-square: $\chi^2(2) = 115.48, p < 0.0001$

that morphological variables are among the most influential variables in predicting the placement of Spanish stress.

5. Discussion

5.1. The experimental results

The experiments reported on in the preceding sections have given us new insight into what cognitive factors (i.e., factors other than acoustic correlates of stress) affect the perception of Spanish stress placement. First, the effect of the stress pattern of a similar word on the perception of stress placement was examined, and it was found that stress perception is affected by the stress pattern of a similar word. Furthermore, the strength of the influence of the real word is gradient, depending on the degree of similarity between the real word and the nonce word. The closer the nonce word is to the real word, the more likely that the stress pattern of the real word will be perceived in the nonce word. It is particularly noteworthy that the degree of similarity goes beyond phonemic similarity, as differing quality between the phonemically identical word-final vowel led to a different degree of influence of the stress pattern of the real word.

Second, the role of syllable weight was reconsidered. A flaw in the experimental design of Face's (2000) study on this topic led to heavy syllables also being longer than light syllables. When this design flaw was corrected in the present study, the results were different. Heavy syllables were found not to attract perceived stress, which is evidence against syllable weight and quantity sensitivity in Spanish. Whether or not the final syllable is open (i.e., ends with the nuclear vowel) or closed (i.e., has a coda consonant) is clearly an important factor in the perception of Spanish stress, but this seems more adequately explained by the nature of the terminal segment than by syllable weight.¹⁴ Either a segmental explanation or a syllable weight explanation is able to account for the effect of the final syllable on stress perception. However, the syllable weight explanation would also predict that heavy syllables elsewhere in the word should attract stress, and the results clearly show that this is not the case. Thus, contrary to Face (2000), it must be concluded that syllable weight does not influence the perception of Spanish stress placement.

Third, it was found that there is an effect of lexical subregularities on the perception of Spanish stress. While the results were not as strong as in a previous production study (Aske 1990), the pattern is clear. These results show that in perception, listeners take into account more segmental material than just the final segment of the word. This adds perceptual

evidence to the production (Aske 1990), computer-modeling (Eddington 2000), and questionnaire (Waltermire 2004) evidence that more than the final segment plays a role in Spanish stress placement.

Finally, the role of morphological category on Spanish stress perception was considered, and it was shown that morphological category has a highly significant effect on stress perception. The perceived stressed syllable depends on whether the nonce word is contextualized as a noun or a verb. This is a significant finding precisely because Aske (1990) found inconclusive results of the effect of morphological category on stress in a nonce word production task. However, the morphological distinction employed in the present study was far more robust than that used in Aske's study, making it clearer to see the effects of morphological category.

While these results provide important information regarding the nature of Spanish stress perception and some of the factors that influence it, another question must be asked: What are the larger implications of these experimental results?

5.2. *Toward a psychological model of speech perception*

Given the different lexical effects on the perception of Spanish stress placement reported in the present study, it is of interest to consider their implications for a psychological model of speech perception. In this section I will briefly describe an exemplar model of speech perception, which seems the most likely platform for a model incorporating the types of lexical effects found in the present study, and indicate how Spanish listeners fill in missing stress placement information based on higher-level abstraction across the space of exemplars. In this way the types of mechanisms that a model would need to incorporate to produce the types of results in the experiment can be considered.

While numerous scholars argue for one version or another of exemplar theory, since the present study deals specifically with speech perception I will limit my description here to the work of Johnson (1997a, b), who has developed a working exemplar model of speech perception. Within Johnson's model, which is illustrated in Figure 1, the first stage of processing involves the conversion of the incoming speech wave into a series of auditory spectra.¹⁵ Each spectrum is compared with a set of spectra stored in the vector quantizing code book, and is coded for which of the stored spectra it is most similar to. If the spectrum in question is not similar to any of the stored spectra, it is added to the codebook. If it is similar to one of the stored spectra, that spectrum is shifted slightly to be more similar to the incoming spectrum. The series of auditory spectra is compared

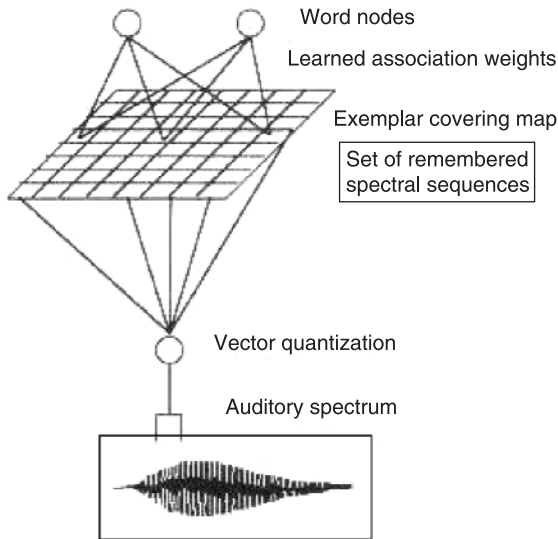


Figure 1. Diagram showing the functioning of Johnson's exemplar model (adapted from Johnson 1997b)

to sequences stored in the exemplar covering map. It is essential here to point out that the exemplar covering map takes the place of actual exemplars (Kruschke 1992), which Johnson takes as an appealing option given that one of the main criticisms of exemplar models is the amount of storage space required. Each location in the covering map corresponds to a set of possible auditory properties, and represents a set of exemplars sharing auditory properties as well as category properties. If the incoming sequence is not similar to any sequence stored in the covering map, the exemplar is added to the map. Attention weights (Nosofsky 1986) control the degree of sensitivity of the process to particular properties, and association weights connect locations on the covering map with word nodes. These association weights are learned as covering map locations are learned to be instances of a particular word.¹⁶

The experimental results of the present study have implications for the characteristics of or the incorporation of three mechanisms within an exemplar model of speech perception. The first of these is the mechanism for comparing the similarity of a lexical item to a stimulus being processed. The similarity of a lexical item to a stimulus being processed is a key component of any exemplar model, but the results of the present study have two implications that both point to differing degrees of activation of exemplars influencing perception. In testing the effects of the stress

pattern of a similar real word on the perception of stress placement in a nonce word, it was shown that replacing the final consonant of the real word with [s], and thereby changing the predicted stress pattern (based on the terminal segment) from final to penultimate, did result in the majority of stimuli being perceived as having penultimate stress. However, final stress was still perceived 37% of the time, which is a far greater percentage than the 13.5% of perceived final stress reported in Face (2000) for words predicted, based on their final segment, to have penultimate stress. Segmentally, once the [s] is perceived in the nonce word it is clear that the listener cannot be processing the similar real word. Nonetheless, the stress pattern of the similar real word does influence the perception of stress placement. An exemplar model, then, must be able to incorporate the influence of similar words even when that word is clearly not being processed. Perhaps the most likely way to accomplish this is for different degrees of activation of exemplars. An exemplar could be activated due to much similarity with the stimulus and thus serve as an analogical model even though that exemplar is not activated to the degree of actually being perceived.

The other implication of the present study for this similarity component is that listeners are sensitive to subphonemic variation. In considering the effects of the stress pattern of a similar real word, the two categories of vowel-final nonce words differed only in the quality of the final vowel. Vowel-final words in Spanish are predicted to have penultimate stress, but were more often perceived as having final stress due to the influence of the stress pattern of the similar real word. However, there was a difference in the degree of influence of the similar real word between the two categories of vowel-final nonce words. For those vowel-final nonce words maintaining the original vowel quality (i.e., that of a vowel in a closed final syllable), final stress was perceived more often, under the influence of the stress pattern of the similar real word, than in those vowel-final nonce words where the same phonemic vowel ended the word but had the quality of a word-final vowel (i.e., in an open final syllable). Subphonemic variation, then, must also be incorporated into models of speech perception, and this again seems plausible by taking into account the degree of activation of exemplars.

The present study also motivates the incorporation of a mechanism of cross-exemplar abstraction. It was found that listeners make use of the final segments of a stimulus for making analogies to patterns in the lexicon. First, when a nonce word ended in a consonant it was generally perceived as having final stress, while when it ended in a vowel it was generally perceived as having penultimate stress. Second, while words ending in [n] were generally perceived as having final stress, there was

more variation for words ending in [en] than for words ending in [an]. Both of these experimental results correspond to patterns observed across the Spanish lexicon. While it can be said that the stimulus is similar in a certain way (e.g., in the nature of its final segments) to numerous words in the Spanish lexicon, this is clearly different from the type of similarity discussed above, which involved a specific exemplar. In this case, the stimulus may not be very similar to any single exemplar and therefore one might not expect to find an analogical influence as no exemplar would be significantly activated. What we are dealing with here is similarity to a pattern found only when patterns across the lexicon are observed. When the entire lexicon is considered, there is a clear correspondence between consonant-final words and final stress, and between vowel-final words and penultimate stress. There is also a clear correspondence between variation in stress placement in words ending in [en] and overwhelmingly final stress in words ending in another vowel followed by [n]. The results of the present study show that listeners compare the stimulus being processed with patterns in the lexicon and use these patterns to make analogies and fill in the missing information for stress placement. The use of this cross-exemplar abstraction of patterns must be incorporated into an exemplar model. Of course such abstraction must be weighted with respect to other factors, as in the case of [s] final words discussed above, where final stress was typically perceived, the general pattern of penultimate stress is outweighed by similarity to a single exemplar that has final stress.

Finally, the results of the present study motivate the incorporation of a mechanism to account for a heavy effect of morphosyntactic categorization. It was found that stress placement was perceived differently for the same stimulus depending on whether it was contextualized as a noun or as a verb. The nonce words ending in [an] in the present study were contextualized as third person present tense verb forms, which always have penultimate stress when multisyllabic, and as nouns, where final stress is the overwhelming pattern in the Spanish lexicon. When these nonce words were contextualized as verbs they were perceived as having penultimate stress, and when they were contextualized as nouns they were perceived as having final stress. Rather than an issue of a single exemplar or a pattern across exemplars influencing perception of an unknown word, the issue raised here is how information other than the exemplars themselves may influence perception.

Johnson (1997a: 147) specifies that “an exemplar is an association between a set of auditory properties and a set of category labels”. These category labels include any classifications that may be of interest to the listener and that was available at the time the exemplar was stored. Johnson gives as examples the linguistic value of the exemplar, the gender of

the speaker, and the name of the speaker. Certainly a part of the linguistic value of the exemplar may be its morphosyntactic function, and Bybee (2001) explicitly lists morphological context among the category information stored in an exemplar. There is certainly evidence for category labels being a part of the information stored in exemplars. Johnson (1997a), for example, uses his exemplar model to account for the fact that listeners correctly identify not only words or utterances produced by different speakers, despite their varying acoustic properties, but also the voices of speakers they have heard before. Johnson et al. (1999: 380) also show that “listeners bring to bear experience-based expectations about talkers in the process of perceiving speech”, and claim that an exemplar-based perceptual system can adequately implement their findings. Thus it seems fairly clear that category labels store diverse kinds of information.

Returning to the issue of the influence of morphological category on the perception of Spanish stress, if morphosyntactic information is stored in exemplars, then this information is available in the lexicon to be used in speech perception. In addition to the acoustic-phonetic properties of the stimulus, a model must allow for morphological context to be able to activate exemplars. Once exemplars are activated, patterns found across the activated exemplars are able to influence perception just as in the case of the other cross-exemplar abstractions discussed above.

6. Conclusion

The present study adds to our knowledge of the cognitive factors that influence the perception of Spanish stress placement and also provides new types of perceptual data that models of speech perception must be able to account for. While Spanish stress is generally correlated to a rise in F0 during the stressed syllable and extended duration of that syllable, other factors influence its perception as well. The present study shows that similar words in the lexicon, the segmental constituency of the final syllable, lexical subregularities, and morphological category all influence the perception of Spanish stress. When the acoustic correlates of stress are neutralized, native Spanish-speaking listeners agree to a large extent on which syllable of a nonce word is stressed, confirming Quilis’s claim that “the sensation of a physical stimulus is channeled through the structures of a language” (1971: 54).

That factors other than acoustic correlates of stress influence the perception of Spanish stress placement provides interesting data to be accounted for by models of speech perception. Given the lexical nature of these factors, an exemplar model seems particularly well suited to account

for them. Yet certain mechanisms must be incorporated into an exemplar model. First, the model must account for the fact that varying degrees of exemplar activation may lead to analogical influence in perception, even when no exemplar is activated to the point of being perceived. Second, it must account for cross-exemplar abstraction of patterns that may influence perception despite a lack of similarity of the stimulus to any single exemplar. Finally, a model must allow for category labels to activate exemplars, which may then trigger the aforementioned cross-exemplar abstraction of patterns.

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Appendix A

List of nonce words and corresponding real words used to test the role of a similar word in Spanish stress perception.

<i>Nonce word</i>	<i>Corresponding real word</i>
arsena(s)	arsenal
calama(s)	calamar
capita(s)	capital/capitán
champiño(s)	champiñón
corazo(s)	corazón
ecuado(s)	Ecuador
funera(s)	funeral
huraca(s)	huracán
hospita(s)	hospital
tenedo(s)	tenedor

Appendix B

List of nonce words used to test the role of syllable weight in Spanish stress perception. Syllable structure is indicated by H (heavy syllable) and L (light syllable).

<i>HHH</i>	<i>HHL</i>	<i>HLH</i>	<i>HLL</i>
bonlandan	bondenda	bandemel	bolnala
dombalden	bonlamba	bondanol	dendana
landangon	dantelda	galdeman	galmeda
lanlendol	malnanga	gondabel	ganloda
menlembal	mandolma	naldelan	landola

<i>LHH</i>	<i>LHL</i>	<i>LLH</i>	<i>LLL</i>
badonguel	banenda	dabonel	beloga
comengon	gadamba	gabadon	dagola
dalandel	gobolda	manaden	dalona
mobalmal	lomenda	noguerol	galema
pelandon	molanga	polanal	mamena

Appendix C

List of nonce words used to test the role of a lexical subregularity on Spanish stress perception. The *-an* group was also used to test the role of morphological category.

<i>-an</i>	<i>-en</i>	<i>-on</i>
balendan	balenden	balendon
cameldan	camelden	cameldon
gamoltan	gamolten	gamolton
gobelman	gobelmen	gobelmon
logolnan	logolnen	logolnon
malamban	malamben	malambon
medolman	medolmen	medolmon
tabonlan	tabonlen	tabonlon
tagandan	taganden	tagandon

Notes

* I am grateful to José Ignacio Hualde, Joan Bybee, and an anonymous reviewer, whose comments on earlier versions led to a much clearer and improved article. Any errors are, of course, my own. Correspondence address: Department of Spanish and Portuguese Studies, University of Minnesota, 51 Folwell Hall, 9 Pleasant Street S. E., Minneapolis, MN 55455, USA. E-mail: face002@umn.edu.

1. While context will certainly serve to disambiguate between segmentally identical words in some cases, the pro-drop nature of Spanish means that an explicit subject will often not be present to provide disambiguating information between two segmentally identical verb forms.
2. An anonymous reviewer mentions that it seems unlikely that stress has no effect on vowel quality in Spanish, noting that if this were the case Spanish would be essentially the same as Japanese in this regard. However, in studies of formant values between stressed and unstressed Spanish vowels (e.g., Quilis. and Esgueva 1983), only minimal and insignificant differences are found.
3. The other purported restriction was that stress must fall on one of the final three syllables of the word. As this falls outside the scope of the discussion of syllable weight, the results of this portion of the study are not discussed here.
4. The MBROLI synthesizer has voices for many different languages, with each created through the compilation of a diphone database.

5. I do not wish to convey that subjects actually believe they heard acoustic cues to stress even though these were removed from the signal. In some cases subjects may believe they really did hear stress, while in other cases their answers may just as likely be a prediction of stress placement. Whichever the case, the experimental results presented in the present study indicate that factors internal to the stimulus (i.e., those controlled in the nonce word task) led to perception of stress placement. It cannot be unequivocally ruled out, however, that external factors played no role.
6. While all speaker/hearers of Spanish are exposed to the acoustic correlates of Spanish stress on a daily basis, explicit identification of them and their function (i.e., as markers of stress) is not a natural use of their knowledge of the Spanish language. While most speaker/hearers are able to accurately identify an acoustically stressed syllable in a nonce word, some do struggle with such a task. In the present study 3 potential subjects were eliminated because they perceived stress correctly less than 90% of the time.
7. One of the nonce words, *capita*, was the beginning of both *capital* 'capital' and *capitán* 'captain'. This is not a problem, however, since both words involve the addition of just one final consonant to the nonce word and both have final stress. Therefore, regardless of the fact that some speakers may take *capita* as similar to *capital* and others may take it as similar to *capitán*, the results would be identical. It was also overlooked that *capita* is the diminutive form of the Spanish noun *capa* 'coating'. Interestingly, however, none of the subjects perceived the penultimate stress of *capita* in this case.
8. José Ignacio Hualde points out that neither maintaining constant vowel durations nor maintaining constant syllable durations is realistic. Rather, Spanish vowels are shorter in heavy syllables than in light syllables, but the syllable itself is longer. For the purposes of the present study, however, it is important that syllable durations be held constant in order to investigate whether phonological syllable weight, independently of syllable duration, affects stress perception. As an anonymous reviewer points out, countering the consonant length by shortening the vowel will actually discourage the perception of stress since the shortened vowels would be too short. This, then, really stacks things against the durational confound in Face (2000).
9. There are verb forms that end in [on], but these are always preceded by [r]. In order to avoid a potential interpretation as such a verb form, none of the nonce words had [r] preceding the final vowel-[n] sequence.
10. An anonymous reviewer points out that to the extent that subjects from different dialectal regions will have different expectations about the acoustic signal, they may be biased differently by choices such as vowel duration patterns in the synthesis. In an attempt to determine if dialectal differences affected the results of the study, the results of this and two previous studies were analyzed. In cases where there were several subjects from the same dialectal region (and therefore reason to believe that individual differences were not the main factor in the results), there are no notable differences between subjects from different dialectal regions.
11. Eddington (2004) and Waltermire (2004) find that syllable weight does influence stress placement, though the difference is likely due to the experimental designs. Eddington performs a computer simulation of Spanish stress placement to investigate potential influencing factors, while Waltermire presents nonce words orthographically and asks subjects to mark stress. Clearly these tasks are quite different from the perception task in the present study and do not represent language processing in real time.
12. That syllables made heavy by being closed with a consonant do not lead to subjects believe they are stressed is in alignment with the findings of Ahn (2000), who shows in a typological study that in unbounded stress systems the only type of weight that attracts stress is that due to vowel length.

13. It is worth noting, however, that even the [en] and [an] sets were much more often perceived as having final stress than as having penultimate stress, indicating that they were most commonly interpreted as not being verbs. In informal tests I have conducted, speakers almost always assign a morphologically ambiguous nonce word the stress pattern it would have as a noun rather than as a verb. While the reasoning for this is beyond the scope of the present study, it is a topic that must be addressed in order to better understand how Spanish speakers process language.
14. While it is the presence or absence of a word-final consonant that determines the most common stress pattern in Spanish, it has been suggested that speakers take into account more segmental information than just the final consonant (Aske 1990; Eddington 2000). In addition, Eddington (2004) finds that the best predictor of Spanish stress involves a combination of terminal segmental information and syllable weight information.
15. In the implementation of his model, Johnson (1997b) employs 23ms frames.
16. This occurs during training of a computer model and presumably as speaker/hearers of a language learn the auditory properties of specific words during language acquisition.

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